Polymath16 now launched: simplifying the lower bound argument for the Hadwiger-Nelson problem

Terence Tao · 14 April 2018

https://terrytao.wordpress.com/2018/04/14/polymath16-now-launched-simplifying-the-lower-bound-argument-for-the-hadwiger-nelson-problem/

tao2typ · converted to Typst 2026-10-14 (UTC)

Just a quick announcement that Dustin Mixon and Aubrey de Grey have just launched the Polymath16 project over at Dustin's blog. The main goal of this project is to simplify the recent proof by Aubrey de Grey that the chromatic number of the unit distance graph of the plane is at least 5, thus making progress on the Hadwiger-Nelson problem. The current proof is computer assisted (in particular it requires one to control the possible 4-colorings of a certain graph with over a thousand vertices), but one of the aims of the project is to reduce the amount of computer assistance needed to verify the proof; already a number of such reductions have been found. See also this blog post where the polymath project was proposed, as well as the wiki page for the project. Non-technical discussion of the project will continue at the proposal blog post.

Comments (3)

 **Anonymous** · 14 April 2018

Dear Terry,sir
May I ask you a little stupid question that you have any works related to Hadwiger-Nelson problem? Thank for your response.

>  **Terence Tao** · 15 April 2018
>
> No, I don't have anything published relating to this problem (I may have written about the tangentially related unit distance problem of Erdos in my book with Van Vu and in some other places, though). I think I played with this problem for a day or so many years ago (hoping to improve upon Falconer's result in the measurable case) but didn't get particularly far. I also assigned the problem of proving CNP ≥ 4 in an undergraduate problem solving class about eighteen years ago; I remember that the majority of students needed quite a few hints to discover the standard Moser spindle solution.

 **Nazgand** · 19 April 2018

Let G_0 be a graph with chromatic number c_0 and vertexes v_a, v_b such that v_a and v_b do not not the same color for any c_0-coloring of G_0. G_0 can then be used as a `virtual edge' with distance $d_0 = \|v_a - v_b\|$. Let the chromatic color of a virtual edge be defined as the chromatic color of the graph used to virtualize the edge.
Multiplicative property of virtual edges:
Let G_1 be a graph with chromatic number c_0 which virtualizes an edge length d_1. Then replacing all edges of G_0 with virtual edges of length d_1 produces a virtual edge of chromatic color c_0 with length d_0*d_1.
If a graph, G_2, with chromatic number $c_2 \geq 1 + c_0$ is created using virtual edges with chromatic number c_0, then a graph, G_3, of chromatic number c_3 (s.t. $c_2 \geq c_3 \geq 1 + c_0$) can be created by replacing the virtual edges with the graphs which virtualize the virtual edges.
More trivially, if a graph, G_4, with chromatic number $c_4 \leq c_0$ is created using virtual edges with chromatic number c_0, then replacing the virtual edges with the graphs which virtualize the virtual edges produces a graph with chromatic number c_0.
If virtual edges with chromatic number 4 are found, then they may be useful to make a graph of chromatic number 5 which uses fewer vertexes than the current smallest graph. In an extreme example, 5 vertices would be enough if each of the distances between pairs of vertices corresponds to a virtual edge with chromatic number 4.